DC1 ↔ DC3 Metro DCI with EVPN Gateways

L2 extension (VLAN 40) + L3 (Type-5) over one EVPN gateway, with DC2 as an L3-only DR site

Item	Detail
Project	Add DC3 as a metro twin of DC1; stretch VLAN 40; route between all three DCs
Lab	EVE-NG on Proxmox; Arista vEOS-lab; DC1 fabric managed by Salt (NAPALM)
Date completed	17 Oct 2026
Result	**All 6 acceptance tests passed** (L2 stretch, L3 both ways, DC2 ↔ DC3 transit)
Key platform note	Border leaves upgraded **EOS 4.31.2F → 4.36.3M** for the full L3 EVPN gateway

1. Goal

Build a third data center (DC3) close to DC1 (metro distance, DWDM) and connect it so that:

1. **VLAN 40 (10.40.40.0/24) exists in both DC1 and DC3**: a host can live in either DC with the same IP and gateway (VM mobility).
2. **DC3 can have its own subnets** (VLAN 70, 10.70.70.0/24) that are routed, not stretched.
3. **All three DCs can reach each other**, including DC2 ↔ DC3 through DC1.
4. Each DC stays its **own EVPN domain**: no DC learns another DC's internal VTEPs, and failures stay local.
5. The design is **future-proof**: stretching another VLAN is a short, repeatable checklist.

2. How it works, in plain language

2.1 The real-world picture

Think of each data center as a **city** with its own postal system.

Network thing	City analogy
Data center (DC1, DC2, DC3)	A city with its own internal postal system
Host (VPC)	A house
VLAN 40	"Main Street", a street that runs through **both** DC1 and DC3
Subnet 10.70.70.0/24 (DC3 only)	"Oak Avenue", a street that exists only in DC3
Anycast gateway 10.40.40.1	The local **post office** on Main Street, one in each city
Border leaf with EVPN gateway	The **customs office** at the edge of each city
Metro DWDM link (Eth5)	The **bridge** between DC1 and DC3
VRF-lite link to DC2 (Eth4.100)	A **toll highway** to a far-away city (DC2)
Type-2 route	A **letter addressed to a person** (exact MAC/IP of a host)
Type-3 route	The **flyer distribution list** (who gets broadcasts on Main Street)
Type-5 route	A **parcel addressed to a ZIP code** (a whole subnet)

2.2 What happens to the mail

- **Letters on Main Street (L2, VLAN 40).** VPC1 in DC1 writes to VPC6 in DC3. DC1 only knows "VPC6 lives behind my customs office". DC1's customs office re-stamps the letter with its own return address and sends it over the bridge; DC3's customs office re-stamps it again and delivers it. Neither city learns the other's street plan.
- **Parcels by ZIP code (L3, Type-5).** VPC7 on Oak Avenue (DC3 only) is reached by ZIP code 10.70.70.0/24. Both customs offices re-stamp these parcels too, so DC1 hosts reach DC3 subnets and vice versa.
- **The far city (DC2).** DC2 is connected by a toll highway that only carries parcels by ZIP code (plain IP routes), never personal letters. DC1's customs office **translates** between the highway and the bridge, so DC2 and DC3 can reach each other through DC1.

- **Local post offices.** Main Street has a post office in **both** cities (anycast gateway 10.40.40.1 on DC1 leaves and on dc3-leaf1), so hosts are always routed locally: no detour (tromboning) to the other DC.

> **Memory hooks**
>
> **Customs re-stamps the mail**: each EVPN gateway re-originates routes with itself as next hop.
>
> **Letters to a name, parcels to a ZIP**: Type-2 = host, Type-5 = subnet.
>
> **Metro pair stretches, DR site routes**: L2 only between nearby DCs; far DCs get L3 only.
>
> **The hub translates**: DC1 border-leaf converts VRF-lite IPv4 ↔ EVPN Type-5.

3. Architecture

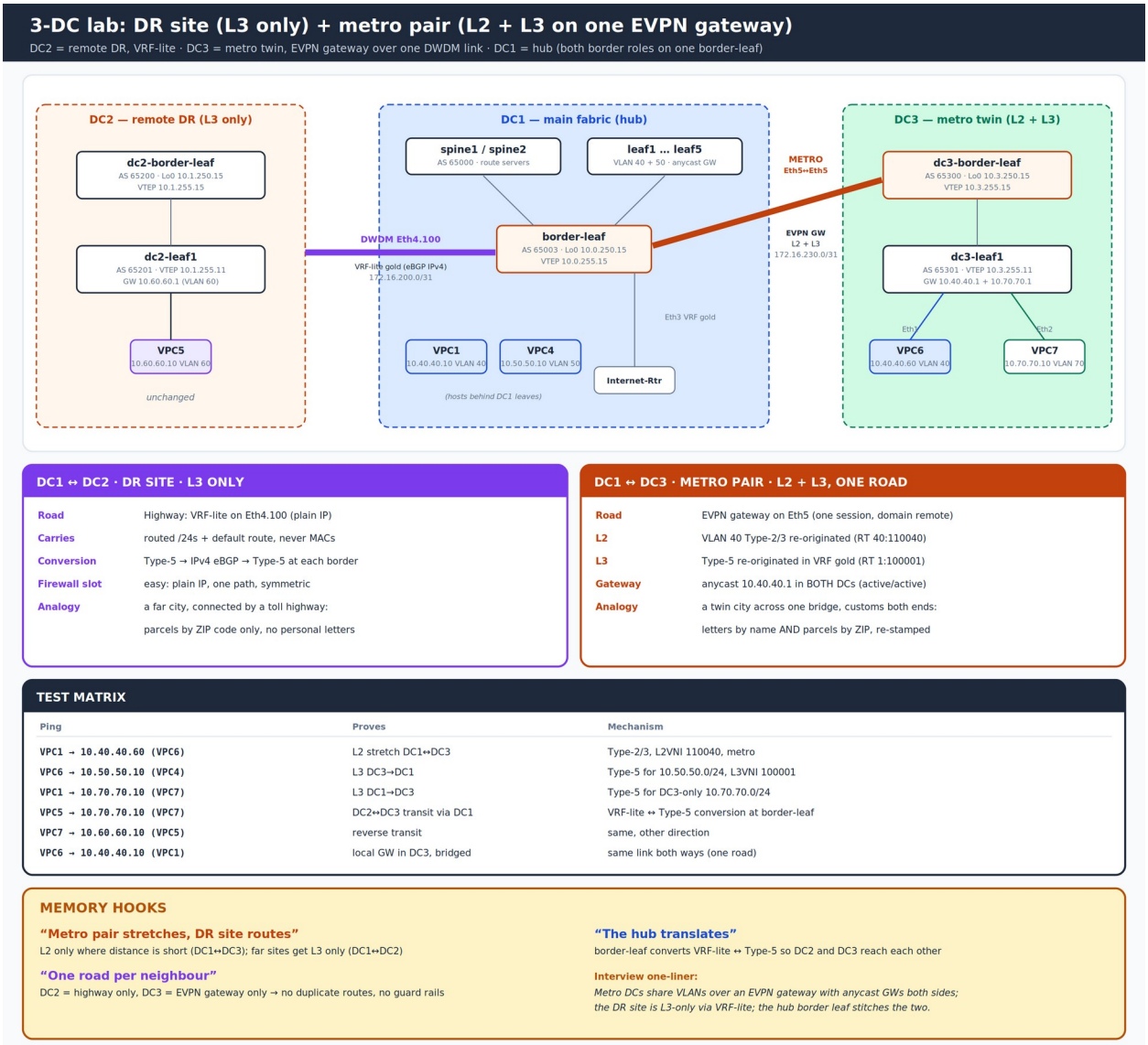


Figure 1: Three-DC design (DC2 = DR, L3 only; DC3 = metro twin, L2 + L3 over one EVPN gateway; DC1 = hub)

3.1 Inventory and addressing

Device	Role	ASN	Lo0 / Lo1 (VTEP)	Key links	EOS
dc1-border-leaf	DC1 EVPN gateway + VRF-lite to DC2 + internet	65003	10.0.250.15 / 10.0.255.15	Eth1/2 spines; Eth3 internet; Eth4.100 DC2; Eth5 DC3	4.36.3M
dc3-border-leaf	DC3 EVPN gateway	65300	10.3.250.15 / 10.3.255.15	Eth5 172.16.230.1/31 (DC1); Eth1 10.3.1.0/31; Lo100 10.3.100.1 (VRF gold)	4.36.3M
dc3-leaf1	DC3 leaf, anycast GW	65301	10.3.250.11 / 10.3.255.11	Eth11 10.3.1.1/31; Eth1 VPC6 (VLAN 40); Eth2 VPC7 (VLAN 70)	4.31.2F
dc2-border-leaf	DC2 border (VRF-lite)	65200	10.1.250.15 / 10.1.255.15	Eth4.100 172.16.200.1/31; Eth1 to dc2-leaf1	4.31.2F

Device	Role	ASN	Lo0 / Lo1 (VTEP)	Key links	EOS
DC1 spines/leaves	DC1 fabric	65000 / 6500x	10.0.250.x / 10.0.255.x	Salt-managed	4.31.2F

Host	Location	VLAN / VNI	IP / gateway	Purpose
VPC1	DC1 leaf1	40 / 110040	10.40.40.10 / .1	Stretched VLAN, DC1 side
VPC4	DC1 leaf3	50 / 110050	10.50.50.10 / .1	DC1-only subnet
VPC5	DC2 dc2-leaf1	60 / 110060	10.60.60.10 / .1	DC2-only subnet
VPC6	DC3 dc3-leaf1 Eth1	40 / 110040	10.40.40.60 / .1	Stretched VLAN, DC3 side
VPC7	DC3 dc3-leaf1 Eth2	70 / 110070	10.70.70.10 / .1	DC3-only subnet (Type-5)

3.2 Key design decisions

Decision	Choice	Why
DC1 ↔ DC3 transport	One EVPN gateway session (domain remote) carrying L2 + L3	One road per neighbour: no duplicate routes, symmetric links
DC1 ↔ DC2 transport	VRF-lite (eBGP IPv4 in VRF gold), unchanged	DR site is L3 only; never stretch L2 over long distance
VLAN 40 gateway	Anycast 10.40.40.1 in both DC1 and DC3	Metro active/active: hosts routed locally, no tromboning
VRF gold RT	Same 1:100001 in all DCs; L3VNI 100001	One tenant VRF end to end
Underlay across DCI	Only the two gateway loopbacks (route-map out)	Neither DC learns the other's internal VTEPs
EOS on border leaves	4.36.3M	4.31 lacks the VRF multi-domain commands (see section 7)
VRF anchor on dc3-border-leaf	Loopback100 in VRF gold	A gateway with no SVIs needs an active VRF to import Type-5

4. Route types: what crosses where

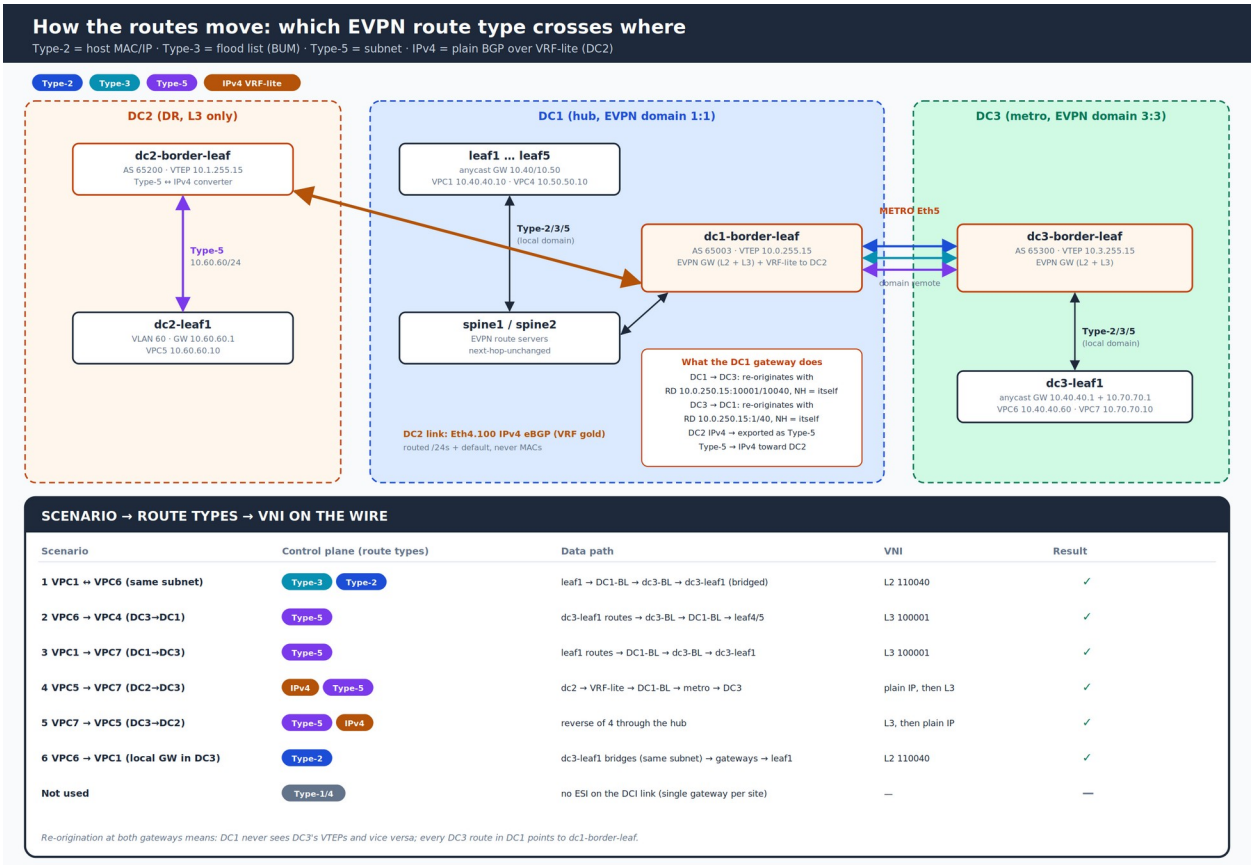


Figure 2: Route types per segment and per test scenario

4.1 EVPN route types used in this project

Type	Name	Carries	Used here for
Type-1	Ethernet Auto-Discovery	ESI + VNI	Not on the DCI (single gateway per site). Used inside DC1 for the leaf4/5 ESI-LAG
Type-2	MAC/IP Advertisement	Host MAC (+IP), L2VNI (+L3VNI)	**VLAN 40 hosts** across DC1 ↔ DC3; host /32 routes for symmetric IRB
Type-3	Inclusive Multicast (IMET)	VTEP IP + L2VNI	**VLAN 40 flood list** (BUM): ends at each gateway, re-originated across
Type-4	Ethernet Segment	ESI + VTEP IP	Not on the DCI. Inside DC1 for leaf4/5 DF election
Type-5	IP Prefix	Subnet + L3VNI + router MAC	**Subnets** 10.40/10.50/10.60/10.70 and the default route between DC1 and DC3
IPv4 (not EVPN)	Plain BGP in VRF gold	Subnet	**DC1 ↔ DC2** over VRF-lite; converted to/from Type-5 at each border leaf

4.2 Scenario summary

#	Test	Route types	Path	Result
1	VPC1 → 10.40.40.60 (VPC6)	Type-3, Type-2	Bridged: leaf1 → DC1-BL → dc3-BL → dc3-leaf1 (L2VNI 110040)	Pass

#	Test	Route types	Path	Result
2	VPC6 → 10.50.50.10 (VPC4)	Type-5	Routed at dc3-leaf1 → dc3-BL → DC1-BL → leaf3/4/5 (L3VNI 100001)	Pass
3	VPC1 → 10.70.70.10 (VPC7)	Type-5	Routed at leaf1 → DC1-BL → dc3-BL → dc3-leaf1 (L3VNI)	Pass
4	VPC5 → 10.70.70.10 (VPC7)	IPv4 then Type-5	dc2 → VRF-lite → DC1-BL → metro EVPN → DC3	Pass
5	VPC7 → 10.60.60.10 (VPC5)	Type-5 then IPv4	Reverse of test 4 through the hub	Pass
6	VPC6 → 10.40.40.10 (VPC1)	Type-2	Same subnet: bridged across both gateways	Pass

5. Configuration walkthrough

Each block below shows the configuration, what every line does, and which route types it creates or affects. DC1 leaves/spines and DC2 were not changed.

5.1 Metro underlay: reach the other gateway's loopbacks

dc1-border-leaf (mirror on dc3-border-leaf with 172.16.230.1 / 10.3.x)

```
interface Ethernet5
   description METRO-EVPN-DCI-to-dc3-border-leaf Ethernet5
   mtu 9214
   no switchport
   ip address 172.16.230.0/31
!
ip prefix-list DCI-UNDERLAY-OUT seq 10 permit 10.0.250.15/32
ip prefix-list DCI-UNDERLAY-OUT seq 20 permit 10.0.255.15/32
route-map DCI-UNDERLAY-OUT permit 10
   match ip address prefix-list DCI-UNDERLAY-OUT
!
router bgp 65003
   neighbor 172.16.230.1 remote-as 65300
   neighbor 172.16.230.1 route-map DCI-UNDERLAY-OUT out
   address-family ipv4
      neighbor 172.16.230.1 activate
```

Line	What it does	Route types
`mtu 9214`	VXLAN adds 50 bytes; jumbo MTU end to end	—
`neighbor 172.16.230.1 remote-as 65300`	eBGP IPv4 session over the DWDM /31	IPv4 (underlay)
`route-map DCI-UNDERLAY-OUT out`	Advertise **only** Lo0 (EVPN peering) and Lo1 (VTEP)	IPv4: 2 prefixes each way

5.2 Metro overlay: the EVPN gateway session

dc1-border-leaf

```
router bgp 65003
   neighbor 10.3.250.15 remote-as 65300
   neighbor 10.3.250.15 update-source Loopback0
   neighbor 10.3.250.15 ebgp-multihop 3
   neighbor 10.3.250.15 send-community extended
   neighbor 10.3.250.15 bfd
   address-family evpn
      domain identifier 1:1
      domain identifier 3:3 remote
      neighbor 10.3.250.15 activate
      neighbor 10.3.250.15 domain remote
```

Line	What it does	Route types
`update-source Loopback0` + `ebgp-multihop 3`	EVPN peering between loopbacks (not the /31)	—

Line	What it does	Route types
`send-community extended`	Carries route-targets, router MAC and VXLAN encap	All EVPN
`domain remote`	**Makes this a gateway session**: routes from DC3 are treated as another domain and are re-originated, not just passed through	Type-2/3/5
`domain identifier 1:1` / `3:3 remote`	Tags each domain (DC1 = 1:1, DC3 = 3:3) for loop prevention	All EVPN
`bfd`	Fast failure detection on the DCI session	—

On dc3-border-leaf the mirror is `domain identifier 3:3` / `domain identifier 1:1 remote` and neighbor `10.0.250.15`.

5.3 L2 gateway: stretching VLAN 40

dc1-border-leaf (dc3-border-leaf identical except RDs 10.3.250.15:40 / :10040)

```
vlan 40
   name APP-40
interface Vxlan1
   vxlan vlan 40 vni 110040
router bgp 65003
   vlan 40
      rd 10.0.250.15:40
      rd evpn domain remote 10.0.250.15:10040
      route-target both 40:110040
      route-target import export evpn domain remote 40:110040
      redistribute learned
```

Line	What it does	Route types
`vxlan vlan 40 vni 110040`	Same L2VNI in both DCs	Type-2/3
`rd 10.0.250.15:40`	RD used when re-originating **into the local domain** (DC1)	Type-2/3
`rd evpn domain remote 10.0.250.15:10040`	RD used when re-originating **toward DC3**	Type-2/3
`route-target both 40:110040`	Import/export with DC1 leaves	Type-2/3
`route-target import export evpn domain remote 40:110040`	Import/export with the DC3 gateway	Type-2/3
`redistribute learned`	Advertise locally learned MACs (none here; gateway only)	Type-2

Result

DC1 leaves see VPC6's MAC behind **10.0.255.15** (dc1-border-leaf). DC3 sees VPC1's MAC behind **10.3.255.15**. Each flood list (Type-3) ends at the local gateway, so BUM traffic crosses the DCI only once.

5.4 L3 gateway: Type-5 between DC1 and DC3 (EOS 4.36)

dc1-border-leaf VRF gold (dc3-border-leaf identical with 10.3.250.15)

```
router bgp 65003
   vrf gold
```

```
      rd 10.0.250.15:1
      rd evpn domain remote 10.0.250.15:10001
      route-target import evpn 1:100001
      route-target export evpn 1:100001
      route-target import evpn domain all 1:100001
      route-target export evpn domain all 1:100001
```

Line	What it does	Route types
`rd 10.0.250.15:1`	RD for VRF routes re-originated into DC1 (local domain)	Type-5
`rd evpn domain remote 10.0.250.15:10001`	RD for VRF routes re-originated **toward DC3**	Type-5
`route-target import/export evpn 1:100001`	Local-domain import/export (DC1 leaves)	Type-5, Type-2 (host /32)
`route-target import/export evpn domain all 1:100001`	**The fix**: import/export in **both** domains, so the gateway installs the other DC's prefixes in its VRF and re-originates them	Type-5

Why this was the turning point

Before this line, DC1 subnets arrived on dc3-border-leaf but were **never installed in its VRF**, so dc3-border-leaf advertised routes it could not forward (VPC6 got "network unreachable" from 10.3.100.1).

After it, dc3-border-leaf shows DC1 prefixes twice: `RD 10.0.250.15:10001 … remote` (received from DC1) and `RD 10.3.250.15:1` (re-originated into DC3 with itself as next hop).

5.5 VRF anchor on dc3-border-leaf

dc3-border-leaf

```
interface Loopback100
   description VRF gold anchor / diagnostic
   vrf gold
   ip address 10.3.100.1/32
router bgp 65300
   vrf gold
      redistribute connected
```

dc3-border-leaf has no SVIs or VRF-lite links in VRF gold. Without an interface the VRF stayed inactive and imported nothing. Loopback100 activates it and doubles as a per-VRF ping target (`ping vrf gold 10.3.100.1`). This mirrors Arista AVD's `vtep_diagnostic` loopback practice. `redistribute connected` advertises it as a Type-5.

5.6 dc3-leaf1: a normal leaf

dc3-leaf1 (key parts)

```
interface Vlan40
   vrf gold
   ip address virtual 10.40.40.1/24
interface Vlan70
   vrf gold
   ip address virtual 10.70.70.1/24
interface Vxlan1
```

```
   vxlan vlan 40 vni 110040
   vxlan vlan 70 vni 110070
   vxlan vrf gold vni 100001
ip virtual-router mac-address 00:1c:73:00:00:99
router bgp 65301
   vlan 40
      rd 10.3.250.11:40
      route-target both 40:110040
      redistribute learned
   vlan 70
      rd 10.3.250.11:70
      route-target both 70:110070
      redistribute learned
   vrf gold
      rd 10.3.250.11:1
      route-target import evpn 1:100001
      route-target export evpn 1:100001
      redistribute connected
```

Piece	What it does	Route types it originates
`ip address virtual` + same vMAC as DC1	Anycast gateway 10.40.40.1: identical IP **and** MAC in both DCs, so a VM keeps its gateway after moving	—
`vlan 40` / `vlan 70` MAC-VRFs	Advertise learned hosts and flood lists	Type-2 (VPC6, VPC7), Type-3 (VLAN 40, 70)
`vrf gold redistribute connected`	Advertise local subnets 10.40.40.0/24 and 10.70.70.0/24	Type-5

dc3-leaf1 knows nothing about DC1 or DC2 VTEPs: every remote route it holds points to **10.3.255.15** (dc3-border-leaf).

5.7 DC2 path (unchanged): VRF-lite translation at the hub

dc1-border-leaf VRF gold (existing)

```
   vrf gold
      neighbor 172.16.200.1 remote-as 65200
      neighbor 172.16.200.1 route-map DCI-OUT out     ! 0.0.0.0/0 + 10.0.0.0/8 le 24
      address-family ipv4
         neighbor 172.16.200.1 activate
```

DC2's 10.60.60.0/24 arrives as **plain IPv4** and is exported into EVPN as **Type-5** (to DC1 leaves and, via the gateway, to DC3). DC3's 10.70.70.0/24 arrives as Type-5 and leaves toward DC2 as IPv4 (it matches `10.0.0.0/8 le 24`). That is why tests 4 and 5 passed with **no change in DC2**.

6. Scenario walkthroughs

Test 1: VPC1 → VPC6 (same subnet, bridged)

- VPC1 ARPs for 10.40.40.60. leaf1 floods the ARP using the **Type-3** list for VNI 110040, which includes dc1-border-leaf.
- dc1-border-leaf re-encapsulates toward dc3-border-leaf (its Type-3 list for the remote domain), which forwards to dc3-leaf1. VPC6 replies.
- dc3-leaf1 advertises VPC6 as **Type-2**; dc3-border-leaf re-originates it to DC1 (NH 10.3.255.15); dc1-border-leaf re-originates it into DC1 (NH 10.0.255.15).
- Unicast: leaf1 → VXLAN 110040 → dc1-border-leaf → VXLAN 110040 → dc3-border-leaf → VXLAN 110040 → dc3-leaf1 (three stitched tunnels).

Test 2: VPC6 → VPC4 10.50.50.10 (DC3 → DC1, routed)

- VPC6 sends to its local gateway 10.40.40.1 on dc3-leaf1.
- dc3-leaf1 looks up VRF gold: 10.50.50.0/24 is a **Type-5** via VTEP 10.3.255.15 (re-originated by dc3-border-leaf), L3VNI 100001.
- dc3-border-leaf looks up its VRF: 10.50.50.0/24 via VTEP 10.0.255.15 (from DC1). dc1-border-leaf looks up its VRF: via leaf3/4/5 (local Type-5 / Type-2 host route).

Test 3: VPC1 → VPC7 10.70.70.10 (DC1 → DC3, routed)

- leaf1 routes in VRF gold: 10.70.70.0/24 **Type-5** via 10.0.255.15 → dc1-border-leaf → dc3-border-leaf → dc3-leaf1 → VPC7 (all on L3VNI 100001).

Test 4/5: VPC5 ↔ VPC7 (DC2 ↔ DC3 transit)

- dc2-leaf1 → dc2-border-leaf (Type-5) → **IPv4** over Eth4.100 → dc1-border-leaf VRF gold → **Type-5** over the metro gateway → dc3-border-leaf → dc3-leaf1. The return reverses it.

Test 6: VPC6 → VPC1 (local gateway in DC3)

- Same subnet, so no routing: dc3-leaf1 bridges using VPC1's **Type-2** (learned via dc3-border-leaf) across the three tunnels.

7. Issues found and how they were solved

#	Symptom	Root cause	Fix
1	`rd evpn domain remote` in VRF → "Ambiguous"; remote RTs → "Invalid input"	EOS 4.31.2F has no multi-domain commands under a VRF (only under MAC-VRFs)	Upgraded border leaves to **4.36.3M**
2	With `next-hop-self received-evpn-routes route-type ip-prefix inter-domain`, only DC1 → DC3 worked	4.31 Type-5 gateway behaviour was one-directional	Replaced by proper VRF multi-domain config on 4.36
3	dc3-border-leaf VRF empty although routes were received	VRF had no interface, so it was inactive	Added **Loopback100** in VRF gold
4	On 4.36, DC1 prefixes still not installed in dc3 VRF	Remote-domain RTs missing (af-ipv4 form not accepted)	`route-target import/export evpn domain all 1:100001` at VRF level

#	Symptom	Root cause	Fix
5	`bgp best-path d-path` rejected	Command not available in this vEOS build	Omitted (optional for two domains)

Lessons learned

The **L2 EVPN gateway** (MAC-VRF `domain remote`) worked on 4.31; the **L3 gateway** needed a newer release. Know your platform's feature level before designing.

A gateway with no local SVIs still needs an **active IP-VRF** (a loopback) to import and re-originate Type-5 routes.

Read the EVPN table and the **VRF routing table** separately: a route can be "valid, best" in EVPN and still not be installed in the VRF.

`show bgp evpn route-type ip-prefix ipv4` showing the remote RD (`10.0.250.15:10001 … remote`) plus a locally re-originated copy is the proof the gateway works.

8. Verification commands

Where	Command	Expected
Both border leaves	`show ip bgp summary`	Spines/leaf + 172.16.230.x Estab
Both border leaves	`show bgp evpn summary`	Local peers + metro peer (domain remote) Estab
dc3-border-leaf	`show bgp evpn route-type ip-prefix ipv4`	DC1 prefixes with RD 10.0.250.15:10001 (remote) and 10.3.250.15:1 (re-originated)
dc3-border-leaf	`show ip route vrf gold`	10.50.50/24, 10.60.60/24, default via VTEP 10.0.255.15
dc3-leaf1	`show ip route vrf gold`	All remote prefixes via VTEP 10.3.255.15
dc1-border-leaf	`show ip route vrf gold 10.70.70.0/24`	Via VTEP 10.3.255.15
leaf1	`show bgp evpn route-type mac-ip 10.40.40.60`	VPC6 with NH 10.0.255.15
dc3-leaf1	`show vxlan address-table vlan 40`	DC1 MACs behind 10.3.255.15
dc2-leaf1	`show ip route vrf gold 10.70.70.0/24`	Via 10.1.255.15 (transit through DC1)

9. Operations

9.1 Stretch another VLAN (example: VLAN 50 / VNI 110050)

1. **dc1-border-leaf:** `vlan 50`, `vxlan vlan 50 vni 110050`, and `router bgp` → `vlan 50` with `rd`, `rd evpn domain remote`, `route-target both 50:110050`, `route-target import export evpn domain remote 50:110050`.
2. **dc3-border-leaf:** the same block with its own RDs.
3. **dc3-leaf1:** `vlan 50`, `vxlan vlan 50 vni 110050`, BGP `vlan 50`, anycast SVI `ip address virtual 10.50.50.1/24`, an access port.
4. **Verify:** Type-2/3 for VNI 110050 on both sides, then ping across.

9.2 Automation status

- DC1 fabric is Salt-managed with `replace: True`. The DCI lines on **dc1-border-leaf are not yet in the templates**: do not run `state.apply` on it until they are, or Salt will remove them. Target others with `salt -L spine1,spine2,leaf1,leaf2,leaf3,leaf4,leaf5 state.apply fabric`.
- Next step: add a `dci:` section (metro peer, stretched VLAN list, VRF multi-domain RD/RT) to the pillar and `leaf.j2`, and onboard DC2/DC3 switches as proxy minions.

9.3 Production hardening (beyond the lab)

Area	Production practice
Redundancy	Two gateways per site with an Interconnect ESI (RFC 9014), all-active
DCI	Two diverse circuits, BFD, MACsec on DWDM
Blast radius	Storm-control and MAC limits on stretched VLANs; approved-VLAN list

Area	Production practice
Firewalls	Stretched subnets + anycast GW cause asymmetric return paths: use a cross-DC firewall cluster with session sync, or zone firewalls at VRF edges
Governance	Stretched VLANs are documented exceptions with an exit date
Monitoring	EVPN routes per VNI, MAC moves (Type-2 sequence numbers), DCI utilisation

Interview one-liner

"DC1 and DC3 are a metro pair joined by EVPN VXLAN gateways (RFC 9014). Each DC is its own EVPN domain; the border leaves re-originate Type-2/3 for the stretched VLAN and Type-5 for subnets with themselves as next hop, so neither DC sees the other's VTEPs. VLAN 40 has an anycast gateway in both DCs. The DR site DC2 is L3-only over VRF-lite, and the hub border leaf translates between VRF-lite and Type-5."